

Bachelor Of Engineering In Information Technology
SECOND YEAR SECOND SEMESTER EXAM – 2024
Subject Name –Graph Theory & Combinatorics (IT/PC/B/T/224)

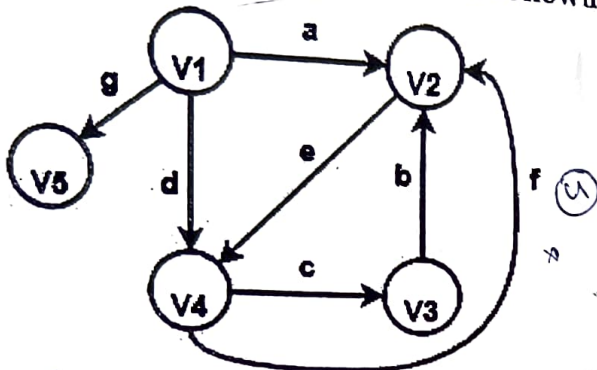
Full Marks=100

CO1

Q1.

[10]

- (i) Suppose G is a non-directed graph with 12 edges. If 6 vertices are each of degree 3 and rest are degree less than 3, what is the minimum number of vertices in G ?
 (ii) Find the circuit matrix of the following graph G .



- (iii) What are the degrees of vertices in a graph G with no adjacent edges?

[4+4+2=10]

CO2

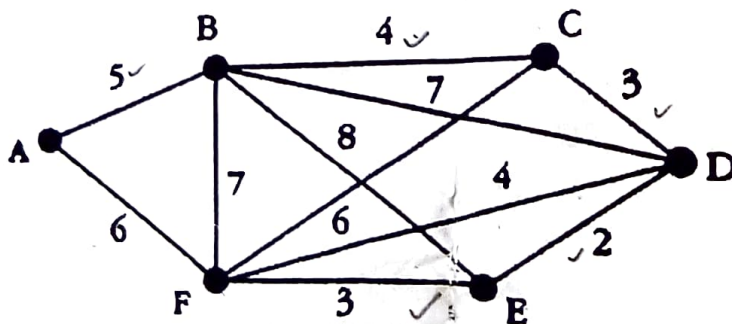
[20]

- Q2. (i) In an electric network consisting of 300 nodes and 500 cables connecting these nodes, find the minimum number of cables that can be removed to reduce the network to a spanning tree. The goal is to remove the cables such that all nodes remain connected without creating isolated groups of nodes.

- (ii) Prove that "Every connected graph has a spanning tree".

- (iii) A rooted-tree has n vertices and k number of pendant vertices. Find the number of vertices having degree 3.

- (iv) Using Kruskal's algorithm find the minimal spanning tree of the following graph.



[4+4+5+7=20]

CO3

Q3.

- (i) Given a graph G with $n = 6$ nodes and $e = 8$ edges, labelled respectively as:

Nodes: A, B, C, D, E, F

Edges: $(A-B), (A-C), (A-D), (B-D), (B-E), (C-F), (D-E), (E-F)$

- a. Identify all the cut edges in the graph, if any and explain your process.

- b. Determine if there are any cut vertices, and if so, identify them.
 c. Determine if there are any minimal cut set that would disconnect node A from node F.
 (ii) Justify this using a suitable example "The vertex connectivity of any graph G can never exceed the edge connectivity of G". (6)

(iii) Given two graphs G and H:

Graph G: $V_1 = \{A, B, C, D, E, F\}$, $E_1 = \{(A, B), (A, C), (B, C), (B, D), (C, E), (D, F), (E, F)\}$

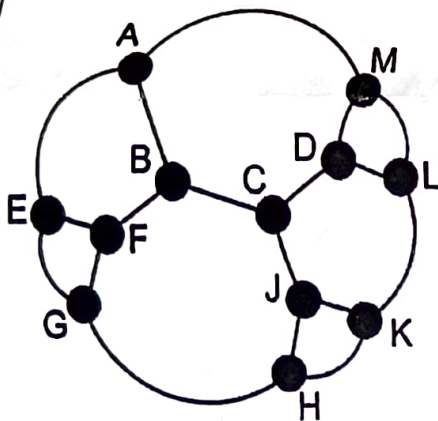
Graph H: $V_2 = \{B, D, E, F, G\}$, $E_2 = \{(B, D), (D, E), (D, F), (E, F), (E, G)\}$

Compute the union of graphs G and H, denoted as $G \cup H$, and provide the resulting set of vertices and edges. [8+6+6=20]

CO4 Q4. (i) "A graph is 2-colorable if it is bipartite and every cycle has an even length."

[20] Justify this with a suitable example. (6)

(ii) Find the **chromatic number** of the graph given below. (Mention all steps properly)



(iii) Consider a bipartite graph G with two sets of vertices:

Set U: $\{A, B, C, D\}$

Set V: $\{1, 2, 3, 4\}$

The graph contains the following edges connecting the two sets:

$(A, 1); (A, 2); (B, 2); (B, 3); (C, 3); (C, 4); (D, 4)$

(a) Find the **maximum matching**, if any, in this bipartite graph.

(b) Find a **maximal matching**, if any, in the graph and list the matching edges.

(c) Determine if the graph has a **perfect matching**. If so, provide the set of edges that form the perfect matching. If not, explain why.

[5+5+(3+3+4)=20]

Q5.

CO5 [20] (i) What is the minimum number of students required in a Graph Theory Class class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

	(ii) Suppose a club has 25 members. How many ways are there to choose a president, vice-president, secretary, and treasurer of the club, where no person can hold more than one post? (iii) Seven women and nine men are on the faculty in the IT department at a University. How many ways are there to select a committee of five members of the department if at least one woman and at least one man must be on the committee? [6+7+7=20]
CO6 Q6. [10]	(i) Find the generating functions of the following sequences in closed form: $\langle 4, 4, 4, 4, 4, \dots \rangle$ (ii) Using generating functions, find a_n in terms of n for the case given below: $a_0 = 1, a_1 = 2 \text{ and } a_{n+2} = 5a_{n+1} - 4a_n \text{ for } n \geq 0$ [5+5=10]

CO1: Explain and discuss the concept of different types of Graphs with fundamental properties and **express** different types of matrix representation. (K2)

CO2: Illustrate different types of trees such as (i) rooted tree (ii) spanning tree etc, and **explain** their properties. (K3)

CO3: Apply operations like Union, Deletion, and decomposition of graphs and **illustrate** Cut vertex and Cut edge and their properties. (K3)

CO4: Illustrate planar graph and their properties and Graph Coloring and Matching. (K3)

CO5: Apply and evaluate basic counting rules, pigeon-hole principle and principle of inclusion-exclusion. (K3)

CO6: Apply and Solve problems using Generating Function and Recurrence Relations. (K3)